

Numerical Analysis homework # 3

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I. Question I

Let $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, then its first derivative would be $p'(x) = a_1 + 2a_2x + 3a_3x^2$, its second derivative would be $p''(x) = 2a_2 + 6a_3x$. And we denote $q(x) = (2 - x)^3$, its first derivative would be $q'(x) = -3(2 - x)^2$, its second derivative would be $q''(x) = 6(2 - x)$. Due to the C^2 continuity condition, we can obtain that $p(1) = q(1)$, that is to say

$$a_0 + a_1 + a_2 + a_3 = 1 \quad (1)$$

$$a_1 + 2a_2 + 3a_3 = -3 \quad (2)$$

$$2a_2 + 6a_3 = 6 \quad (3)$$

And it easy to grasp that $p(0) = a_0 = q(0) = 0$. By equation(1)(2)(3), we can obtain that $a_1 = 12, a_2 = -18, a_3 = 7$, which implies that $p(x)$ satisfies

$$p(x) = 12x - 18x^2 + 7x^3 \quad (4)$$

By the definition of natural cubic spline, we have $s''(0) = s''(2)$. However, by computing we can obtain that $s''(2) = 0$, while $s''(0) = -36 \neq 0$, which is contrary to the definition of natural cubic spline, so $s(x)$ is not a natural cubic spline.

II. Question II

II.a

In the $n - 1$ interval $[x_i, x_{i+1}]$, the spline $s(x)$ is a quadratic polynomial, we denote it as $p_i(x) = a_ix^2 + b_ix + c_i$. Since there are $n - 1$ intervals and 3 unknown coefficients for each interval, thus we have a total of $3(n - 1)$ unknown parameters. While the definition of $s \in S_2^1$ provides the following constraints:

1. The spline must pass through all the data points, that is, $s(x_i) = f_i$ holds for $i = 1, 2, \dots, n$
2. The polynomial pieces must join continuously at the $n - 2$ internal knots, which can give us $n - 2$ conditions, that is $p_i(x_{i+1}) = p_{i+1}(x_{i+1})$ holds for $i = 1, 2, \dots, n - 2$
3. The first derivative must also be continuous at the $n - 2$ internal knots, which can give us another $n - 2$ conditions, that is, $p'_i(x_{i+1}) = p'_{i+1}(x_{i+1})$ for $i = 1, 2, \dots, n - 2$

To conclude, we have $3n - 4$ constraints, while we have $3n - 3$ unknown parameters, so the constraints is not enough to solve the unknown parameters. Therefore, an additional condition is needed to determine s uniquely.

II.b

In other words, we need to find the specific quadratic polynomial $p_i(x)$ on the interval $[x_i, x_{i+1}]$ given three pieces of Information:

1. $p_i(x_i) = s(x_i) = f_i$

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2. $p_i(x_{i+1}) = s(x_{i+1}) = f_{i+1}$
3. $p'_i(x_i) = s'(x_i) = m_i$

We can use Newton form to to construct this polynomial centered at x_i , that is to say

$$p_i(x) = C_0 + C_1(x - x_i) + C_2(x - x_i)^2 \quad (5)$$

Then the derivative of $p_i(x)$ is $p'_i(x) = C_1 + 2C_2(x - x_i)$. Since $p_i(x_i) = f_i, p'_i(x_i) = m_i$, we can obtain that $C_0 = f_i, C_1 = m_i$, then $p_i(x)$ can be written as

$$p_i(x) = f_i + m_i(x - x_i) + C_2(x - x_i)^2 \quad (6)$$

In addition, we have $p_i(x_{i+1}) = f_i + m_i(x_{i+1} - x_i) + C_2(x_{i+1} - x_i)^2 = f_{i+1}$, we can obtain that C_2 satisfies

$$C_2 = \frac{f_{i+1} - f_i - m_i(x_{i+1} - x_i)}{(x_{i+1} - x_i)^2} \quad (7)$$

Finally, we have

$$p_i(x) = f_i + m_i(x - x_i) + \frac{f_{i+1} - f_i - m_i(x_{i+1} - x_i)}{(x_{i+1} - x_i)^2}(x - x_i)^2 \quad (8)$$

II.c

By definition, we have $m_{i+1} = s'(x_{i+1}) = p'_{i+1}(x_{i+1})$, due to the C^1 continuity, we have $m_{i+1} = p'_i(x_{i+1})$, and we can use the conclusion of II.b, that is equation(8), we have

$$m_{i+1} = p'_i(x_{i+1}) = m_i + 2\frac{f_{i+1} - f_i}{x_{i+1} - x_i} - 2m_i \quad (9)$$

The equation can be turned into a recursive formula

$$m_{i+1} = 2\frac{f_{i+1} - f_i}{x_{i+1} - x_i} - m_i \quad (10)$$

Since $m_1 = f'(a)$ is given, we can compute m_2, m_3, m_{n-1} recursively by equation(10)

III. Question III

Let $s_2(x)$ be $s_2(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, then its first derivative and second derivative would be $s'_2(x) = a_1 + 2a_2x + 3a_3x^2, s''_2(x) = 2a_2 + 6a_3x$, and the first, second derivative of $s_1(x)$ is $s'_1(x) = 3c(x+1)^2, s''_1(x) = 6c(x+1)$. Since $s(x)$ is a natural cubic spline on $[-1, 1]$, and a natural spline must have C^2 continuity at all internal knots and must have zero second derivatives at the endpoints, while $s''(-1) = 0$ is automatically satisfied. So we try to satisfy the C^0, C^1, C^2 continuity, we can obtain that

- $s_1(0) = 1 + c = s_2(0) = a_0$
- $s'_1(0) = 3c = s'_2(0) = a_1$
- $s''_1(0) = 6c = s''_2(0) = 2a_2$
- $s''(1) = s''_2(1) = 2a_2 + 6a_3 = 0$

Thus, we have $a_0 = c - 1, a_1 = 3c, a_2 = 3c, a_3 = -c$, then $s_2(x)$ can be written as $s_2(x) = 1 + c + 3cx + 3cx^2 - cx^3$. Furthermore, we have $s(1) = 1 + c + 3c + 3c - c = -1$, from which we can obtain that $c = -\frac{1}{3}$

IV. Question IV

IV.a

We have $f_0 = f(-1) = \cos(-\frac{\pi}{2}) = 0, f_1 = f(0) = \cos(0) = 1, f_2 = f(1) = \cos(\frac{\pi}{2}) = 0$. And we denote $s(x)$ is defined by $s_0(x)$ on $[-1, 0]$, $s_1(x)$ on $[0, 1]$, while $M_i = s''(x_i)$ is the second derivatives at the knots. Then the natural boundary conditions would be $M_0 = s''(-1) = 0, M_2 = s''(1) = 0$, and the knot spacing would be

$h_0 = x_1 - x_0 = 1, h_1 = x_2 - x_1 = 1$ We can use the standard system of equations for cubic splines. For the single internal knot $x_1 = 0$, we can obtain that

$$h_0 M_0 + 2(h_0 + h_1)M_1 + h_1 M_2 = 6\left(\frac{f_2 - f_1}{h_1} - \frac{h_1 - f_0}{h_0}\right) \quad (11)$$

By substituting h_0, h_1, M_0, M_2 , we can obtain that $M_1 = -3$, thus, the second derivatives at the knots are $(M_0, M_1, M_2) = (0, -3, 0)$. Then, we can use the general formula for a spline segments $s_i(x)$ on the interval $[x_i, x_{i+1}]$, that is to say

$$s_i(x) = M_i \frac{(x_{i+1} - x)^3}{6h_i} + M_{i+1} \frac{(x - x_i)^3}{6h_i} + \left(\frac{f_i}{h_i} - \frac{M_i h_i}{6}\right)(x_{i+1} - x_i) + \left(\frac{f_{i+1}}{h_i} - \frac{M_{i+1} h_i}{6}\right)(x - x_i) \quad (12)$$

Then for $s_0(x)$ on $[-1, 0]$, we have $x_i = -1, x_{i+1} = 0, h_i = 1, f_i = 0, f_{i+1} = 1, M_i = 0, M_{i+1} = -3$, by substituting into equation(12), we can obtain that

$$s_0(x) = -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 1 \quad (13)$$

For $s_1(x)$ on $[0, 1]$, we have $x_i = 0, x_{i+1} = 1, h_i = 1, f_i = 1, f_{i+1} = 0, M_i = -3, M_{i+1} = 0$, by substituting into equation(12), we can obtain that

$$s_1(x) = \frac{1}{2}x^3 - \frac{3}{2}x^2 + 1 \quad (14)$$

Thus, we can get a natural cubic spline interpolant

$$s(x) = \begin{cases} -\frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, & x \in [-1, 0] \\ \frac{1}{2}x^3 - \frac{3}{2}x^2 + 1, & x \in [0, 1] \end{cases} \quad (15)$$

IV.b

$g(x)$ is the quadratic polynomial that interpolates f at $(-1, 0), (0, 1), (1, 0)$, we denote $g(x)$ as $g(x) = a + bx + cx^2$. Then we have

$$g(0) = a = 1 \quad (16)$$

$$g(1) = a + b + c = 0 \quad (17)$$

$$g(-1) = a - b + c = 0 \quad (18)$$

Then we can obtain that $c = -1, b = 0, a = 1$, that is, $g(x) = 1 - x^2$.

To calculate $E(s)$, we need to find the second derivatives of $s(x)$, we have $s'_0(x) = -\frac{3}{2}x^2 - 3x, s''_0(x) = -3x - 3, s'_1(x) = \frac{3}{2}x^2 - 3x, s''_1(x) = 3x - 3$, then we can integrate $[s''(x)]^2$ over the full interval:

$$E(s) = \int_{-1}^0 [s''_0(x)]^2 dx + \int_0^1 [s''_1(x)]^2 dx = 6 \quad (19)$$

To calculate $E(g)$, we have

$$E(g) = \int_{-1}^1 [g''(x)]^2 dx = \int_{-1}^1 (-2)^2 dx = 8 \quad (20)$$

To calculate $E(f)$, we have

$$E(f) = \int_{-1}^1 [f''(x)]^2 dx = \int_{-1}^1 \left(-\frac{\pi^2}{4} \cos\left(\frac{\pi}{2}x\right)\right)^2 dx = \frac{\pi^4}{16} \quad (21)$$

While $E(f) = \frac{\pi^4}{16} \approx 6.088$, it is easy to obtain that $E(s) < E(f), E(s) < E(g)$, the minimal energy property is verified.

V. Question V

V.a

We can use the Cox-de Boor formula. The $B_i^2(x)$ spline is built from $B_i^1(x)$ and $B_{i+1}^1(x)$. The recurrence relation is

$$B_i^2(x) = \frac{x - t_{i-1}}{t_{i+1} - t_{i-1}} B_i^1(x) + \frac{t_{i+2} - x}{t_{i+2} - t_i} B_{i+1}^1(x) \quad (22)$$

And the basic functions $B_i^1(x)$ and $B_{i+1}^1(x)$ are defined as follows:

$$B_i^1(x) = \begin{cases} \frac{x-t_{i-1}}{t_i-t_{i-1}}, & x \in [t_{i-1}, t_i] \\ \frac{t_{i+1}-x}{t_{i+1}-t_i}, & x \in [t_i, t_{i+1}] \\ 0 & \text{otherwise} \end{cases} \quad (23)$$

$$B_{i+1}^1(x) = \begin{cases} \frac{x-t_i}{t_{i+1}-t_i}, & x \in [t_i, t_{i+1}] \\ \frac{t_{i+2}-x}{t_{i+2}-t_{i+1}}, & x \in [t_{i+1}, t_{i+2}] \\ 0 & \text{otherwise} \end{cases} \quad (24)$$

Thus, we can derive $B_i^2(x)$ by combining these over its full support $[t_{i-1}, t_{i+2}]$.

For $x \in [t_{i-1}, t_i]$, we have

$$B_{i+1}^1(x) = 0, \quad (25)$$

and

$$B_i^2(x) = \frac{x-t_{i-1}}{t_{i+1}-t_{i-1}} \left(\frac{x-t_{i-1}}{t_i-t_{i-1}} \right) \quad (26)$$

For $x \in [t_i, t_{i+1}]$, we have

$$B_i^2(x) = \frac{x-t_{i-1}}{t_{i+1}-t_{i-1}} \left(\frac{t_{i+1}-x}{t_{i+1}-t_i} \right) + \frac{t_{i+2}-x}{t_{i+2}-t_i} \left(\frac{x-t_i}{t_{i+1}-t_i} \right) \quad (27)$$

For $x \in [t_{i+1}, t_{i+2}]$, we have

$$B_i^2(x) = 0 + \frac{t_{i+2}-x}{t_{i+2}-t_i} \left(\frac{t_{i+2}-x}{t_{i+2}-t_{i+1}} \right) \quad (28)$$

Otherwise, $B_i^2(x) = 0$

V.b

First, we can compute the derivative $\frac{d}{dx} B_i^2(x)$ for each piece, for simplicity, we denote $w_i = t_{i+1} - t_i$.

For $x \in (t_{i-1}, t_i)$, $\frac{d}{dx} B_i^2(x) = \frac{2(x-t_{i-1})}{w_i(t_i-t_{i-1})}$;

For $x \in (t_i, t_{i+1})$, $\frac{d}{dx} B_i^2(x) = \frac{d}{dx} \left[\frac{(x-t_{i-1})(t_{i+1}-x)}{w_i(t_{i+1}-t_i)} \right] = \frac{t_{i+1}+t_{i-1}-2x}{w_i(t_{i+1}-t_i)} + \frac{t_{i+2}+t_i-2x}{w_{i+1}(t_{i+1}-t_i)}$;

For $x \in (t_{i+1}, t_{i+2})$, $\frac{d}{dx} B_i^2(x) = \frac{-2(t_{i+2}-x)}{w_{i+1}(t_{i+2}-t_{i+1})}$;

Then it is easy to obtain that at knot t_i , we have:

$$\lim_{x \rightarrow t_i^-} \frac{d}{dx} B_i^2(x) = \frac{2(t_i-t_{i-1})}{w_i(t_i-t_{i-1})} = \frac{2}{t_{i+1}-t_{i-1}} \quad (29)$$

$$\lim_{x \rightarrow t_i^+} \frac{d}{dx} B_i^2(x) = \frac{t_{i+1}+t_{i-1}-2t_i}{w_i(t_{i+1}-t_i)} + \frac{t_{i+2}+t_i-2t_i}{w_{i+1}(t_{i+1}-t_i)} = \frac{2}{t_{i+1}-t_{i-1}} \quad (30)$$

$$(31)$$

So the derivative is continuous at t_i .

At knot t_{i+1} , we have:

$$\lim_{x \rightarrow t_{i+1}^-} \frac{d}{dx} B_i^2(x) = \frac{-t_{i+1}+t_{i-1}}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{t_{i+2}+t_i-2t_{i+1}}{w_{i+1}(t_{i+1}-t_i)} = \frac{-2}{t_{i+2}-t_i} \quad (32)$$

$$\lim_{x \rightarrow t_{i+1}^+} \frac{d}{dx} B_i^2(x) = \frac{-2(t_{i+2}-t_{i+1})}{w_{i+1}(t_{i+2}-t_{i+1})} = \frac{-2}{t_{i+2}-t_i} \quad (33)$$

So the derivative is continuous at t_{i+1} .

V.c

From V.b, we have $\frac{d}{dx} B_i^2(x) = \frac{2(x-t_{i-1})}{w_i(t_i-t_{i-1})}$ when $x \in (t_{i-1}, t_i)$, so $\frac{d}{dx} B_i^2(x) > 0$ holds for all $x \in (t_{i-1}, t_i)$, which means the function is increasing.

And we have $\frac{d}{dx} B_i^2(x) = \frac{-2(t_{i+2}-x)}{w_{i+1}(t_{i+2}-t_{i+1})}$ when $x \in (t_{i+1}, t_{i+2})$, so $\frac{d}{dx} B_i^2(x) < 0$ holds for all $x \in (t_{i+1}, t_{i+2})$, which means the function is decreasing.

Since the function is increasing in (t_{i-1}, t_i) and decreasing at (t_{i+1}, t_{i+2}) , then the single critical point x^* must lie in the middle interval (t_i, t_{i+1}) . Let the derivative on x^* be zero, we can obtain that

$$\frac{t_{i+1} + t_{i-1} - 2x^*}{t_{i+1} - t_{i-1}} + \frac{t_{i+2} + t_i - 2x^*}{t_{i+2} - t_i} = 0 \quad (34)$$

from which we can solve x^* , and x^* satisfies

$$x^* = \frac{t_{i+1}t_{i+2} - t_it_{i-1}}{t_{i+2} + t_{i+1} - t_i - t_{i-1}} \quad (35)$$

V.d

First of all, we prove that $B_i^2(x) \geq 0$. By equation(26)(27)(28), it is easy to obtain that $B_i^2(x) \geq 0$, because each factor is non-negative.

Then we prove that $B_i^2(x) \leq 1$, we can use the B-splines $B_i^k(x)$'s partition of unity, that is $\sum_i B_i^k(x) = 1$ for all x in the domain. Since $\sum_i B_i^2(x) = 1$, and $B_i^2(x) \geq 0$ holds, then it is easy to obtain $B_i^2(x) \leq 1$ holds.

Therefore, $B_i^2(x) \in [0, 1]$

V.e

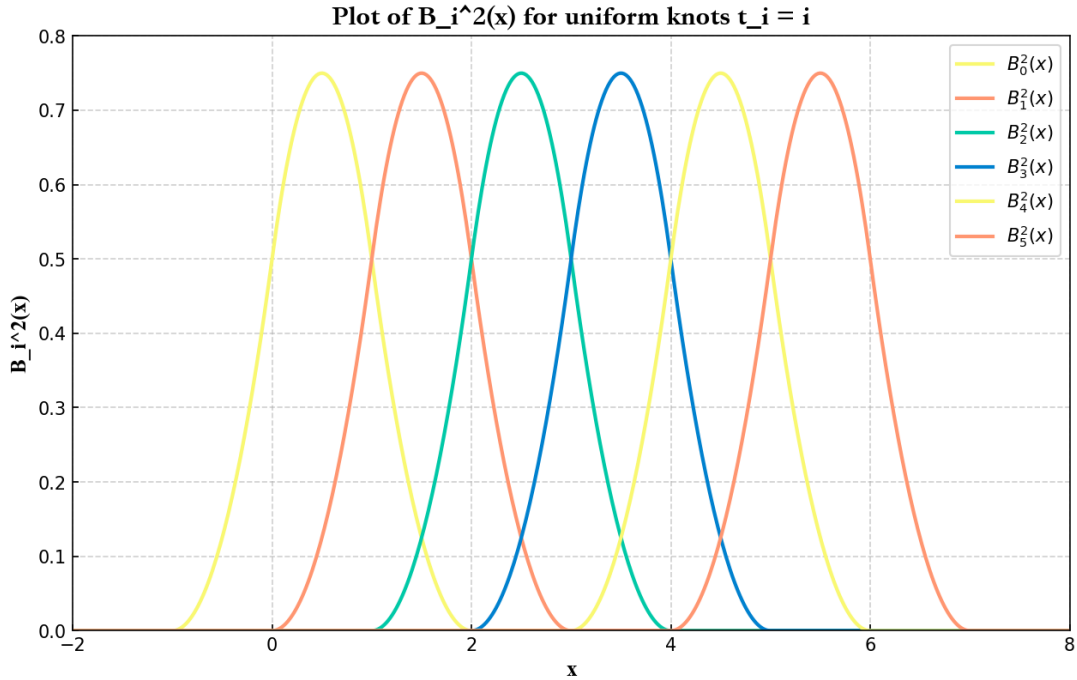


Figure 1: Plot of $B_i^2(x)$ for uniform knots $t_i = i$.

This plot illustrates a series of quadratic B-spline basis functions, $B_i^2(x)$, under the condition uniform knots $t_i = i$

VI. Question VI

Let $f(t) = (t - x)_+^2$, LHS can be expanded using the definition of divided differences:

$$LHS = (t_{i+2} - t_{i-1}) \frac{[t_i, t_{i+1}, t_{i+2}]f - [t_{i-1}, t_i, t_{i+1}]f}{t_{i+2} - t_{i-1}} = [t_i, t_{i+1}, t_{i+2}]f - [t_{i-1}, t_i, t_{i+1}]f \quad (36)$$

And RHS is the definition of $B_i^2(x)$ as we derived in V.a4paper The problem can be divided into several cases as below:

- Case1: $x \geq t_{i+2}$, then for all $t \in \{t_{i-1}, t_i, t_{i+1}, t_{i+2}\}$, we have $t \leq t_{i+2} \leq x$, which means $f(t) = (t - x)_+^2 = 0$ holds for all knots. Then we have LHS=RHS=0

- Case2: $x < t_{i-1}$, then for all $t \in \{t_{i-1}, t_i, t_{i+1}, t_{i+2}\}$, we have $t \geq t_{i-1} > x$, which means $f(t) = (t-x)_+^2 = (t-x)^2$, while the 3-order divided differences of any quadratic is zero, that is, LHS = 0, while $B_i^2(x) = 0$ holds too, that is, RHS = 0, so we have LHS = RHS
- Case3: $x \in [t_{i+1}, t_{i+2})$, then we have $f(t_{i-1}) = 0, f(t_i) = 0, f(t_{i+1}) = 0, f(t_{i+2}) = (t_{i+2} - x)^2$, then we consider LHS $[t_i, t_{i+1}, t_{i+2}]f - [t_{i-1}, t_i, t_{i+1}]f$, we can obtain that

$$[t_{i-1}, t_i, t_{i+1}]f = \frac{[t_i, t_{i+1}]f - [t_{i-1}, t_i]f}{t_{i+1} - t_{i-1}} = 0 \quad (37)$$

$$[t_i, t_{i+1}, t_{i+2}]f = \frac{[t_{i+1}, t_{i+2}]f - [t_i, t_{i+1}]f}{t_{i+2} - t_i} = \frac{\frac{f(t_{i+2}) - f(t_{i+1})}{t_{i+2} - t_{i+1}} - 0}{t_{i+2} - t_i} = \frac{(t_{i+2} - x)^2}{(t_{i+2} - t_i)(t_{i+2} - t_{i+1})} \quad (38)$$

Which means LHS = $\frac{(t_{i+2}-x)^2}{(t_{i+2}-t_i)(t_{i+2}-t_{i+1})}$ = RHS

- Case4: $x \in [t_{i-1}, t_i)$, then we have $f(t_{i-1}) = 0, f(t_i) = (t_i - x)^2, f(t_{i+1}) = (t_{i+1} - x)^2, f(t_{i+2}) = (t_{i+2} - x)^2$, then we consider LHS $[t_i, t_{i+1}, t_{i+2}]f - [t_{i-1}, t_i, t_{i+1}]f$, we can obtain that

$$[t_i, t_{i+1}, t_{i+2}]f = 1 \quad (39)$$

$$[t_{i-1}, t_i, t_{i+1}]f = \frac{t_{i+1} + t_i - 2x - \frac{(t_i - x)^2}{t_i - t_{i-1}}}{t_{i+1} - t_{i-1}} \quad (40)$$

Thus, by simplification, we can obtain that LHS = $\frac{(x-t_{i-1})^2}{(t_{i+1}-t_{i-1})(t_i-t_{i-1})}$ = RHS

- Case5: $x \in [t_i, t_{i+1})$, we have $f(t_{i-1}) = 0, f(t_i) = 0, f(t_{i+1}) = (t_{i+1} - x)^2, f(t_{i+2}) = (t_{i+2} - x)^2$, then we consider LHS $[t_i, t_{i+1}, t_{i+2}]f - [t_{i-1}, t_i, t_{i+1}]f$, we can obtain that

$$[t_{i-1}, t_i, t_{i+1}]f = \frac{[t_i, t_{i+1}]f - [t_{i-1}, t_i]f}{t_{i+1} - t_{i-1}} = \frac{(t_{i+1} - x)^2}{(t_{i+1} - t_{i-1})(t_{i+1} - t_i)} \quad (41)$$

$$[t_i, t_{i+1}, t_{i+2}]f = \frac{[t_{i+1}, t_{i+2}]f - [t_i, t_{i+1}]f}{t_{i+2} - t_i} = \frac{\frac{(t_{i+2}-x)^2 - (t_{i+1}-x)^2}{t_{i+2} - t_{i+1}} - \frac{(t_{i+1}-x)^2}{t_{i+1} - t_i}}{t_{i+2} - t_i} \quad (42)$$

by simplification, we have LHS = $\frac{(x-t_{i-1})(t_{i+1}-x)}{(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} + \frac{(t_{i+2}-x)(x-t_i)}{(t_{i+2}-t_i)(t_{i+1}-t_i)} = \text{RHS}$

To conclude, the equation holds true for all possible intervals of x .

VII. Question VII

According to Theorem 3.34 of the book, we have

$$\frac{d}{dx} B_i^{n+1}(x) = (n+1) \left(\frac{B_i^n(x)}{t_{i+n}-t_{i-1}} - \frac{B_{i+1}^n(x)}{t_{i+n+1}-t_i} \right) \quad (43)$$

Then we can obtain that

$$\int_{t_{i-1}}^{t_{i+n+1}} \frac{d}{dx} B_i^{n+1}(x) dx = (n+1) \int_{t_{i-1}}^{t_{i+n+1}} \left(\frac{B_i^n(x)}{t_{i+n}-t_{i-1}} - \frac{B_{i+1}^n(x)}{t_{i+n+1}-t_i} \right) dx \quad (44)$$

LHS = $[B_i^{n+1}(x)]_{t_{i-1}}^{t_{i+n+1}} = B_i^{n+1}(t_{i+n+1}) - B_i^{n+1}(t_{i-1}) = 0$. Since LHS = 0, we have to prove RHS=0, which is equivalent to

$$\int_{t_{i-1}}^{t_{i+n+1}} \frac{B_i^n(x)}{t_{i+n}-t_{i-1}} dx = \int_{t_{i-1}}^{t_{i+n+1}} \frac{B_{i+1}^n(x)}{t_{i+n+1}-t_i} dx \quad (45)$$

And we need to be aware that the support of $B_i^n(x)$ is $[t_{i-1}, t_{i+n}]$, and the support of $B_{i+1}^n(x)$ is $[t_i, t_{i+n+1}]$. so we just need to consider the integration in their own supports, that is:

$$\int_{t_{i-1}}^{t_{i+n}} \frac{B_i^n(x)}{t_{i+n}-t_{i-1}} dx = \int_{t_i}^{t_{i+n+1}} \frac{B_{i+1}^n(x)}{t_{i+n+1}-t_i} dx \quad (46)$$

which means that the scaled integral of $B_i^n(x)$ is identical to the scaled integral of $B_{i+1}^n(x)$. By recursion, we can prove that the value of the scaled integral is constant for all indices i .

VIII. Question VIII

VIII.a

By definition, we can obtain that $\tau_2(x_i, x_{i+1}, x_{i+2})$ satisfies

$$\tau_2(x, x_{i+1}, x_{i+2}) = x_i^2 + x_{i+1}^2 + x_{i+2}^2 + x_i x_{i+1} + x_i x_{i+2} + x_{i+1} x_{i+2} \quad (47)$$

And then we can compute $[x_i, x_{i+1}, x_{i+2}]x^4$, we have

$$[x_i, x_{i+1}]x^4 = \frac{x_{i+1}^4 - x_i^4}{x_{i+1} - x_i} = x_{i+1}^3 + x_{i+1}^2 x_i + x_{i+1} x_i^2 + x_i^3 \quad (48)$$

$$[x_{i+1}, x_{i+2}]x^4 = \frac{x_{i+2}^4 - x_{i+1}^4}{x_{i+2} - x_{i+1}} = x_{i+2}^3 + x_{i+2}^2 x_{i+1} + x_{i+2} x_{i+1}^2 + x_{i+1}^3 \quad (49)$$

Then we can obtain that

$$[x_i, x_{i+1}, x_{i+2}]x^4 = \frac{[x_{i+1}, x_{i+2}]x^4 - [x_i, x_{i+1}]x^4}{x_{i+2} - x_i} = x_{i+2}^2 + x_{i+2} x_i + x_i^2 + x_{i+1} x_{i+2} + x_{i+1} x_i + x_{i+1}^2 \quad (50)$$

Thus, we have verified LHS=RHS

VIII.b

We can prove the theorem by induction. By Lemma 3.45 of the book, we have

$$\tau_k(x_i, \dots, x_{i+n}) = \frac{\tau_{k+1}(x_{i+1}, \dots, x_{i+n}) - \tau_{k+1}(x_i, \dots, x_{i+n-1})}{x_{i+n} - x_i} \quad (51)$$

Let $P(n)$ be the proposition $[x_i, \dots, x_{i+n}]x^m = \tau_{m-1}(x_i, \dots, x_{i+n})$.

When $n = 0$, $[x_i]x^m = x_i^m = \tau_m(x_i)$, the base case holds.

And we assume $P(k)$ is true for $n = k$, where $k < m$, that is to say, for any k knots, we have

$$[x_i, \dots, x_{i+k}]x^m = \tau_{m-k}(x_i, \dots, x_{i+k}) \quad (52)$$

Then we prove $P(k+1)$ is true, that is, $[x_i, \dots, x_{i+k+1}]x^m = \tau_{m-k-1}(x_i, \dots, x_{i+k+1})$ Through Lemma 3.45, we can obtain that

$$\tau_{m-(k+1)}(x_i, \dots, x_{i+k+1}) = \frac{\tau_{m-k}(x_{i+1}, \dots, x_{i+k+1}) - \tau_{m-k}(x_i, \dots, x_{i+k})}{x_{i+k+1} - x_i} \quad (53)$$

Then by applying the inductive hypothesis, we have $\tau_{m-k}(x_{i+1}, \dots, x_{i+k+1}) = [x_{i+1}, \dots, x_{i+k+1}]x^m$ and $\tau_{m-k}(x_i, \dots, x_{i+k}) = [x_i, \dots, x_{i+k}]x^m$ Then by substituting we have

$$\tau_{m-(k+1)}(x_i, \dots, x_{i+k+1}) = \frac{[x_{i+1}, \dots, x_{i+k+1}]x^m - [x_i, \dots, x_{i+k}]x^m}{x_{i+k+1} - x_i} \quad (54)$$

Therefore, we have $\tau_{m-(k+1)}(x_i, \dots, x_{i+k+1}) = [x_i, \dots, x_{i+k+1}]x^m$. So by induction, the theorem holds for all $n = 0, 1, \dots, m$